

AUTHORIZED SECURITY LAB DOSSIER

Task 02 — Legal Scope, Asset Inventory & Threat Model

RabTech Academy — Authorized Security Lab

Prepared for cybersecurity training and assessment

Document status: Draft for submission review

Prepared date: 02 September 2026

1. Purpose and Engagement Context

This dossier defines the authorization boundary, approved assets, testing restrictions, evidence-handling requirements, threat model, and rules of engagement for the authorized security lab used for the RabTech training exercise. The purpose is to establish a controlled and auditable basis for security testing before reconnaissance, validation, or other assessment activities are performed.

Testing is restricted to systems explicitly included in the approved asset inventory. No system outside the inventory is considered authorized merely because it is reachable from the student's virtual machine or local network.

2. Written Authorization

The engagement owner is identified as RabTech lab in the supplied Rules of Engagement. The approved testing assets are LAB-01 and LAB-02. The assessor is recorded as the student assessor for this training exercise. The formal approval date, testing window, and emergency contact must be completed from the lab's approved engagement record rather than invented by the assessor.

Authorization Field	Recorded Value
Lab owner	RabTech lab
Assessor	Student assessor — enter assigned name
Approval date	
Engagement purpose	Authorized cybersecurity training and security assessment
Approved assets	LAB-01 and LAB-02 only
Emergency contact	
Testing window	Student-defined approved window; enter exact start and end time
Evidence classification	Confidential training evidence

2.1 Authorization Boundary

The authorization boundary is limited to the localhost training web application and the deliberately vulnerable virtual machine identified in the asset inventory. Third-party systems, public Internet targets, unrelated virtual machines, host operating-system services, and other networks are outside the engagement boundary unless they are explicitly added to the inventory and approved.

3. Authorized Asset Inventory

The following inventory reproduces the supplied engagement scope. The values in the inventory are the authoritative scope values for this dossier.

Asset ID	Asset Name	Owner	Environment	IP / URL	Allowed	Exclusions	Evidence
LAB-01	Local training web app	RabTech lab	Isolated localhost	http://127.0.0.1:8080	Yes	No denial of service or third-party scanning	Confidential training evidence
LAB-02	Deliberately vulnerable VM	RabTech lab	Private host-only network	192.168.56.20	Yes	No persistence or outbound traffic	Confidential training evidence

3.1 Asset Address Reconciliation

During lab setup, the deliberately vulnerable VM was observed using the address 192.168.56.128, while the supplied authorization inventory identifies LAB-02 as 192.168.56.20. This is a material scope discrepancy. The observed address must not be silently substituted for the authorized inventory value. Before any further testing is

treated as authorized, the lab owner or assessor should confirm that 192.168.56.128 is the same approved LAB-02 asset and update the inventory if required.

The connectivity evidence and reconnaissance evidence already collected against 192.168.56.128 should therefore be retained as lab evidence but clearly marked as performed against the observed address pending scope reconciliation. This avoids falsely representing an unlisted address as authorized.

4. Rules of Engagement

4.1 Permitted Activities

Permitted activity consists of controlled reconnaissance, validation, evidence collection, and reporting against approved assets inside the isolated lab. Reconnaissance may include host discovery where appropriate, TCP service and port identification, service/version enumeration, and documentation of exposed attack surfaces. Validation must remain within the approved techniques and must not cross the defined exclusions.

For LAB-01, testing is restricted to the localhost training application at 127.0.0.1:8080. For LAB-02, testing is restricted to the deliberately vulnerable virtual machine at the inventory-approved address after the address discrepancy has been resolved.

4.2 Explicit Exclusions

Denial-of-service activity, destructive payloads, persistence mechanisms, credential reuse, third-party targets, and uncontrolled data extraction are prohibited. Outbound traffic from LAB-02 is also excluded. Any action that could affect systems outside the approved lab boundary must not be performed.

4.3 Testing Window and Stop Conditions

Testing shall occur only during the approved window recorded by the lab owner. The assessment must stop immediately if an asset is found to be outside the inventory, if traffic leaves the isolated lab, if a system shows signs of instability, if evidence handling requirements cannot be maintained, or if the emergency contact instructs the assessor to stop.

Control	Requirement
Start time	[Enter approved start time]
End time	[Enter approved end time]
Emergency contact	[Enter contact]
Rate limits	Use conservative scan rates appropriate for the lab; stop if instability appears
Immediate stop condition	Out-of-scope asset, outbound traffic, instability, destructive effect, or lab-owner instruction

4.4 Evidence Handling

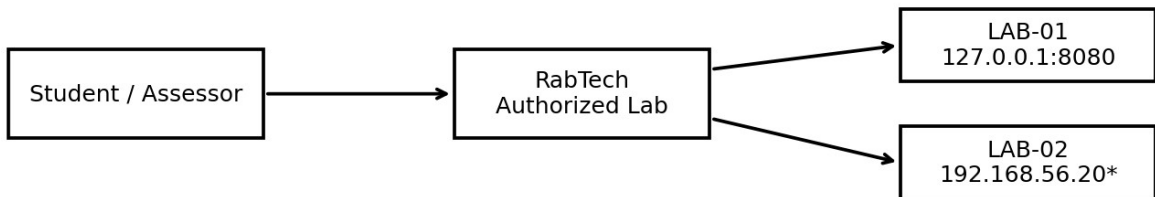
Evidence shall be limited to the minimum material required to demonstrate a finding. Screenshots, logs, command output, and notes must be stored as confidential training evidence. Secrets, credentials, tokens, personal data, and unrelated system information must be redacted where possible. Evidence should be deleted after the retention period specified by the lab or training program; the exact deletion date should be recorded in the final submission.

4.5 Reporting Requirement

Each security finding must identify the affected asset, provide a severity rationale, describe the reproduction or validation method, explain the impact, identify the supporting evidence, and provide remediation guidance. Findings must distinguish confirmed observations from assumptions or unverified possibilities.

5. Data-Flow Diagram

The following diagram represents the intended assessment boundary and the flow of assessment activity from the student assessor to the authorized RabTech lab assets. LAB-02 is shown using the supplied inventory address and is explicitly marked for reconciliation with the observed lab address.



*Inventory value supplied in the RoE. Actual lab addressing must be reconciled before testing.

Figure 1. Authorized assessment data-flow and scope boundary.

6. STRIDE Threat Model

The STRIDE register below maps relevant abuse cases to the authorized lab environment. The ranking is intended to prioritize defensive risk rather than authorize exploitation.

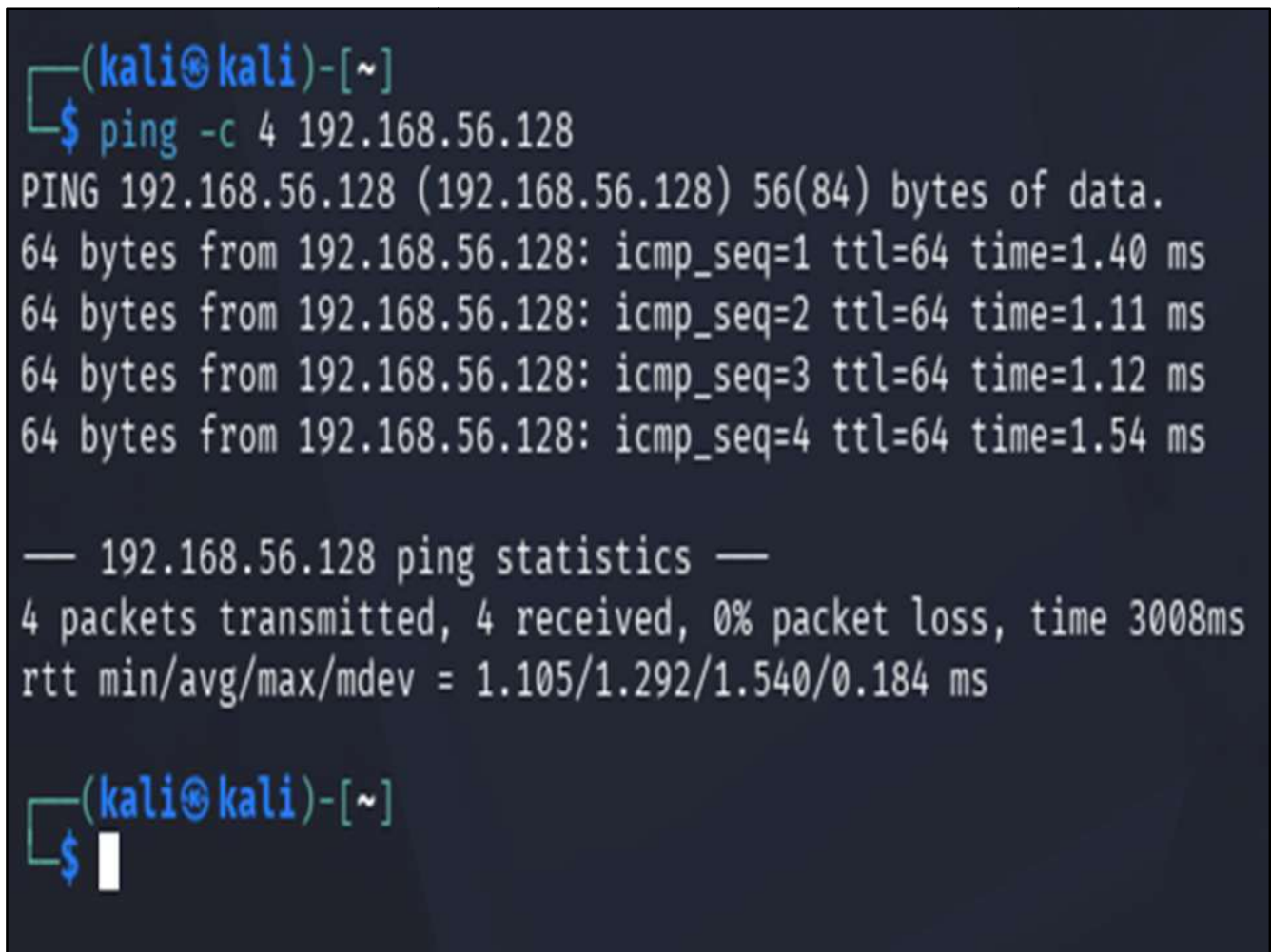
STRIDE Category	Relevant Abuse Case	Asset / Surface	Risk Rationale	Priority Mitigation
Spoofing	Unauthorized impersonation of a lab account or service identity	LAB-01 / LAB-02 authentication surfaces	Weak or reused credentials could allow unauthorized access within the lab.	Use unique lab credentials and restrict authentication exposure.
Tampering	Unauthorized modification of application or system data	LAB-01 web application; LAB-02 services	Changes could alter evidence or the behavior of the deliberately vulnerable system.	Restrict write access and maintain clean snapshots/backups.
Repudiation	Inability to attribute an action to an assessor or account	Lab logs and assessment evidence	Poor attribution weakens auditability and incident reconstruction.	Preserve relevant timestamps, commands, and service logs.
Information Disclosure	Exposure of confidential training data, service data, or credentials	LAB-01 / LAB-02 and collected evidence	Uncontrolled extraction violates the engagement boundary and can expose secrets.	Collect minimum evidence and redact secrets and personal data.
Denial of Service	Service disruption caused by aggressive or destructive testing	LAB-01 / LAB-02 services	Disruption can invalidate the lab and is explicitly prohibited by the RoE.	No DoS testing; use conservative rates and immediate stop conditions.
Elevation of Privilege	Obtaining privileges beyond the authorized testing account	LAB-02 privileged services	Privilege escalation could change system state and exceed the intended validation boundary.	Keep validation controlled and prohibit persistence or destructive actions.

7. Evidence and Screenshot Register

The screenshot register provides a place for the assessor to insert evidence captured during the engagement. Screenshots should show enough context to establish the command, target, result, and timestamp where possible.

Evidence ID	Suggested File Name	Evidence Purpose	Insert Below
SS-01	SS-01-kali-metasploitable-connectivity.png	Demonstrates network reachability between the assessment workstation and the observed lab VM.	Screenshot placeholder
SS-02	SS-02-nmap-service-scan.png	Shows initial service and version discovery against the observed lab VM.	Screenshot placeholder
SS-03	SS-03-nmap-full-port-scan.png	Shows the complete TCP port inventory observed on the lab VM.	Screenshot placeholder
SS-04	SS-04-nmap-detailed-services.png	Shows detailed service enumeration and standard discovery results.	Screenshot placeholder

7.1 Screenshot SS-01



```

(kali㉿kali)-[~]
$ ping -c 4 192.168.56.128
PING 192.168.56.128 (192.168.56.128) 56(84) bytes of data.
64 bytes from 192.168.56.128: icmp_seq=1 ttl=64 time=1.40 ms
64 bytes from 192.168.56.128: icmp_seq=2 ttl=64 time=1.11 ms
64 bytes from 192.168.56.128: icmp_seq=3 ttl=64 time=1.12 ms
64 bytes from 192.168.56.128: icmp_seq=4 ttl=64 time=1.54 ms

— 192.168.56.128 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 1.105/1.292/1.540/0.184 ms

(kali㉿kali)-[~]
$

```

Purpose: Network reachability evidence. Paste the captured ping screenshot here.

7.2 Screenshot SS-02

```
(kali@kali)-[~]
$ sudo nmap -Pn -sV 192.168.56.128
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-02 08:42 -0400
Nmap scan report for 192.168.56.128
Host is up (0.0022s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 57.22 seconds

(kali@kali)-[~]
$
```

Purpose: Initial service/version enumeration evidence. Paste the captured Nmap screenshot here.

7.3 Screenshot SS-03

```
(kali@kali)-[~]
$ sudo nmap -Pn -p- 192.168.56.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-02 08:46 -0400
Nmap scan report for 192.168.56.128
Host is up (0.0010s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
32828/tcp open  unknown
41456/tcp open  unknown
42554/tcp open  unknown
51498/tcp open  unknown
MAC Address: 00:0C:29:FA:DD:2A (VMware)

Nmap done: 1 IP address (1 host up) scanned in 11.46 seconds
```

Purpose: Full TCP port inventory evidence. Paste the captured Nmap screenshot here.

7.4 Screenshot SS-04

```

$ sudo nmap -Pn -sV -sC -p 21,22,23,25,53,80,111,139,445,512,513,514,1099,1524,2049,2121,3306,3632,5432,5900,6000,6667,6697,8009,8180,8787,32828,41456,42554,51498 192.168.56.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-02 08:52 -0400
Nmap scan report for 192.168.56.128
Host is up (0.0019s latency).

PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 192.168.56.129
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_smtp-command: metasplitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|_program version port/proto service
|_100003 2,3,4 2049/tcp nfs
|_100003 2,3,4 2049/udp nfs
|_100005 1,2,3 50193/udp mountd
|_100005 1,2,3 51498/tcp mountd
|_100021 1,3,4 32828/tcp nlockmgr
|_100021 1,3,4 45333/udp nlockmgr
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
143/tcp   open  ntp         Samba smbd 3.X - 4.X (workgroup: WORKGROUP)

```

Purpose: Detailed service enumeration evidence. Paste the captured Nmap screenshot here.

8. Scope Decision and Assessment Readiness

The engagement is ready for controlled assessment only after the authorization record is complete and the LAB-02 address discrepancy is resolved. The supplied inventory identifies 192.168.56.20, whereas the lab VM observed during setup used 192.168.56.128. Because the Rules of Engagement explicitly prohibit testing systems absent from the asset inventory, the observed address requires confirmation or an inventory update before it can be treated as an approved target.

This dossier therefore records the scope exactly as supplied, documents the discrepancy instead of concealing it, and provides a controlled basis for subsequent reconnaissance and validation.

9. Approval and Sign-Off

Role	Name	Date	Signature / Approval
Lab Owner	RABTECH LAB		
Student Assessor	PRITAM BOSE	02-09-2026	/S/ Pritam Bose
Training Reviewer			

Document control note: Before final submission, replace all bracketed placeholders with the approved engagement information and confirm that the LAB-02 IP address is consistent with the authorized inventory.